

Introducing StockSense

AI-Powered NSE Stock Analysis



Presented by the StockSense Team

The Problem

Bridging the investment knowledge gap

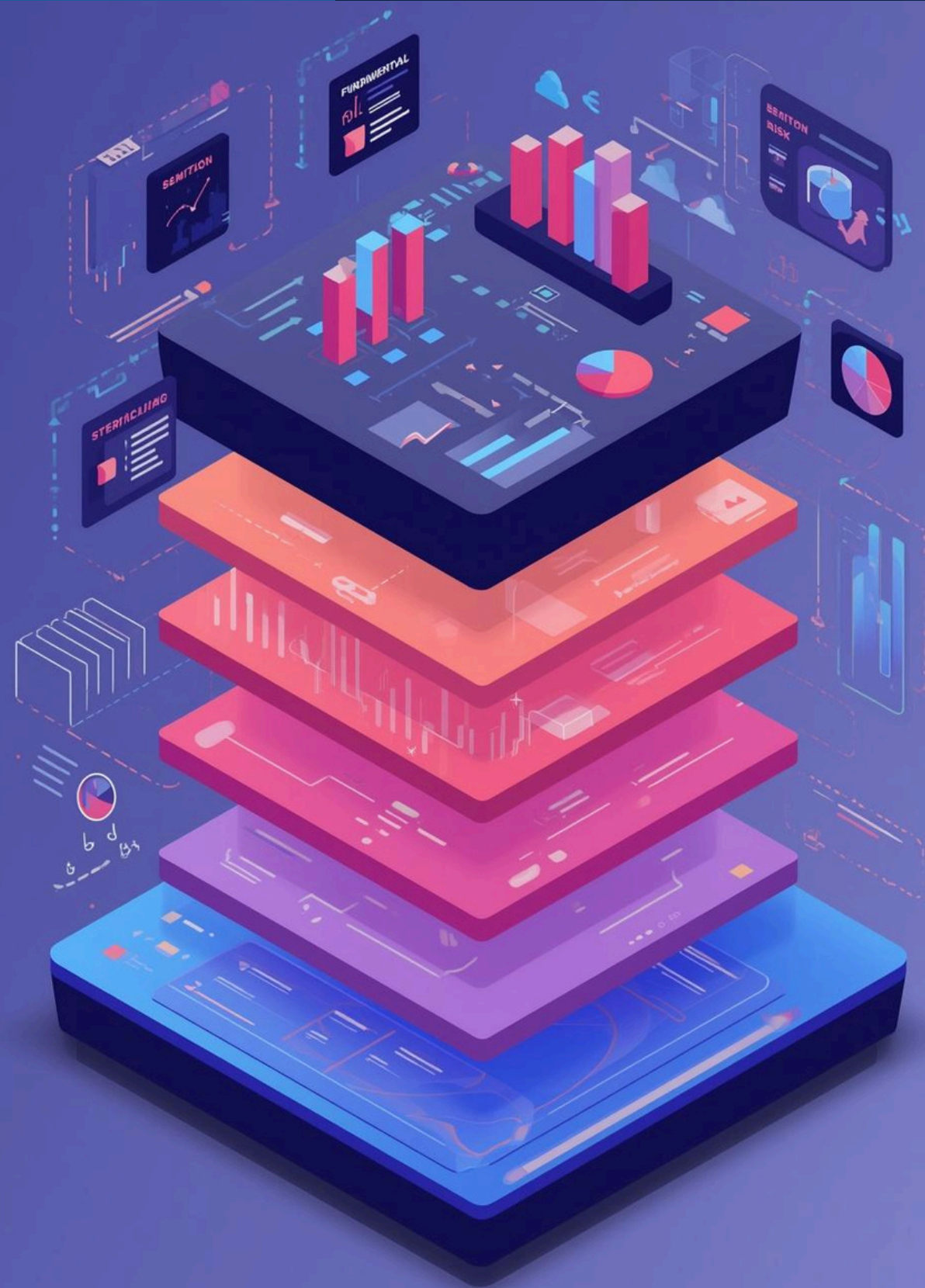
Empowering Retail Investors

Retail investors often lack access to **professional-grade tools** for stock analysis. StockSense addresses this gap by providing AI-driven signals and ratings, making informed investing accessible to everyone.

5-Layer Framework

Analyzing Stocks Effectively

The **5-layer analysis framework** enhances stock evaluation by integrating multiple metrics, ensuring detailed insights across fundamental, technical, sentiment, risk, and AI scoring dimensions for informed investment decisions.



Fundamental Analysis

Key Metrics for Stock Evaluation

Fundamental analysis is crucial for evaluating the intrinsic value of a stock. It focuses on key financial metrics that investors need to assess a company's overall health and performance. The primary metrics considered in this analysis include:

- **PE Ratio:** Indicates how much investors are willing to pay for each rupee of earnings.
- **EPS Growth:** Measures the company's profitability over time.
- **Debt-Equity Ratio:** Assesses the financial leverage and risk level.
- **Free Cash Flow:** Represents the cash available for distribution to investors.
- **Return on Equity (ROE):** Evaluates the efficiency in generating profits from shareholders' equity.

Each of these metrics contributes to 35% of the final stock score, providing investors with a comprehensive view of a stock's fundamental strength.

Technical Analysis

Analyzing Price Momentum and Trends

Technical analysis is a crucial component of the StockSense platform, as it helps investors understand price movements and market trends.

Key indicators used in this layer include:

- **Relative Strength Index (RSI)**: Measures the speed and change of price movements.
- **Moving Average Convergence Divergence (MACD)**: Identifies bullish or bearish momentum.
- **Bollinger Bands**: Assesses market volatility and potential price reversals.
- **Simple Moving Averages (SMA)**: Tracks average price over specific periods, with 50 and 200-day SMAs being particularly significant.

Each of these indicators plays a vital role in determining overall market conditions. By integrating these calculations into our 5-layer analysis framework, StockSense provides users with a comprehensive understanding of stock performance, helping them make informed investment decisions based on calculated momentum and trend directions.

Sentiment & Macro

Analyzing Market Influences and Trends

Understanding market sentiment is critical for predicting stock performance. StockSense leverages advanced Natural Language Processing (NLP) to score news headlines and assess their impact on investor sentiment.

Key elements of the sentiment analysis include:

- **News Headline Scoring:** Evaluates the positivity or negativity of news articles related to stocks.
- **Institutional Flow Tracking:** Monitors Foreign Institutional Investor (FII) and Domestic Institutional Investor (DII) activities.
- **RBI Policy Impact:** Assesses the implications of Reserve Bank of India (RBI) decisions on market trends.
- **Sectoral Momentum:** Analyzes the performance of various sectors to gauge overall market sentiment.

By scoring these factors, StockSense provides retail investors with actionable insights, allowing for well-informed buying, holding, or avoiding decisions. This multifaceted approach ensures that users receive a comprehensive view of market trends, enhancing their investment strategies.

Risk Scoring

Quantifying Downside Risk Effectively

The Risk Scoring layer evaluates multiple critical factors to determine the potential downside risk of stocks. Here's a breakdown of the key components:

- **Beta:** Measures the stock's volatility relative to the market.
- **Maximum Drawdown:** Assesses the largest observed loss from a peak to a trough.
- **Sharpe Ratio:** Evaluates risk-adjusted returns, indicating performance relative to risk.
- **52-Week Price Positioning:** Analyzes the current price relative to its range over the past year.

Each of these elements contributes to a comprehensive understanding of risk, being weighted to reflect their importance in the final score. By employing this risk scoring system, investors can confidently gauge potential threats, thus making informed decisions based on robust data, ultimately enhancing their investment strategies in the dynamic NSE environment.

AI Signal Engine

Integrating Data for Stock Insights

The **AI Signal Engine** is the core of StockSense's analytical capabilities. It synthesizes the outputs from the four preceding layers:

- Fundamental Analysis (Weight: 35)
- Technical Analysis (Weight: 30)
- Sentiment Analysis (Weight: 15)
- Risk Scoring (Weight: 20)

Using a weighted formula, the engine produces a unified score between 0 and 100. This score categorizes stocks into three actionable signals:

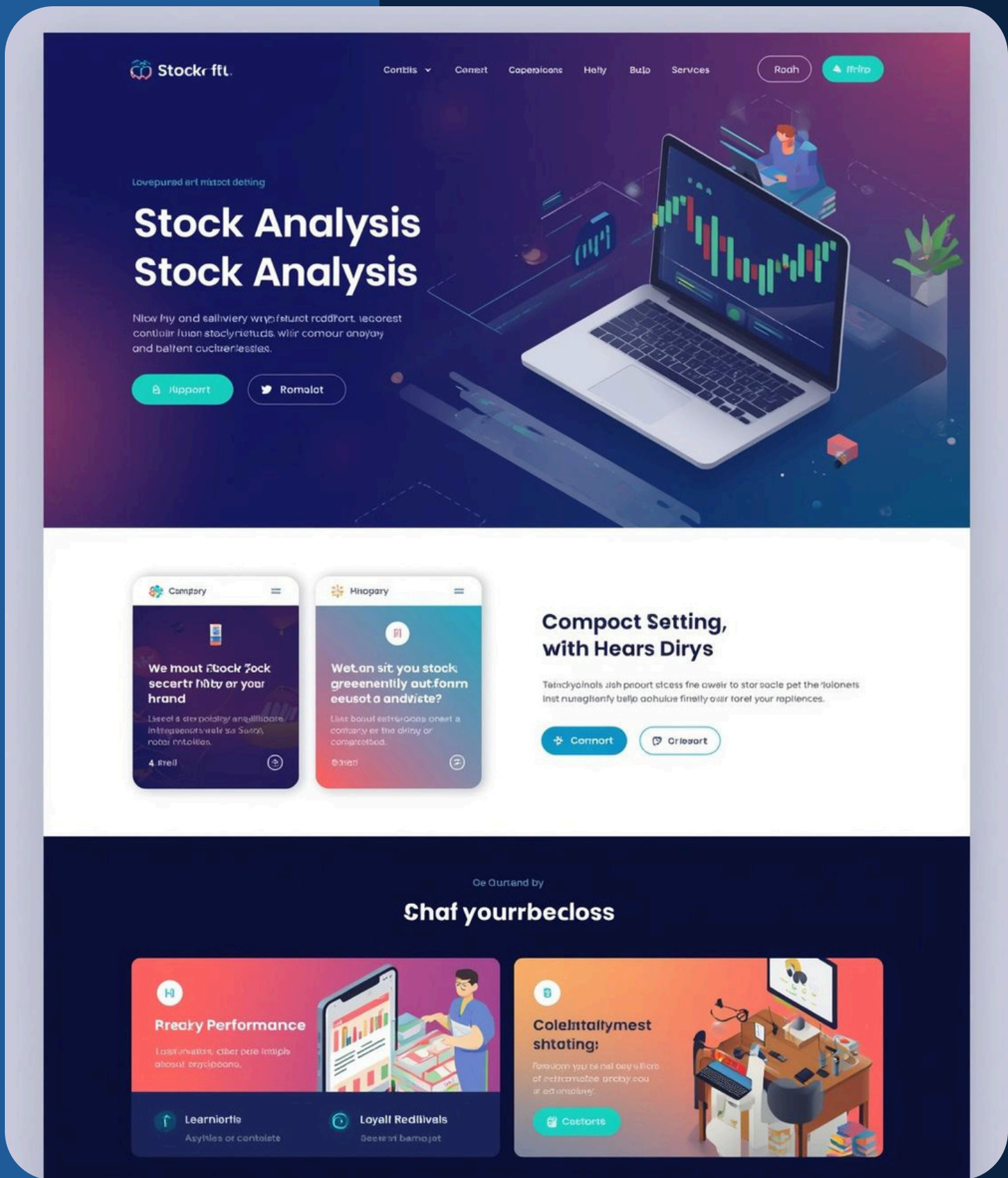
- **BUY:** Score of 65 or above
- **HOLD:** Score between 42 and 64
- **AVOID:** Score below 42

By implementing advanced algorithms and machine learning techniques, StockSense aims for an impressive target accuracy of over 80% in its evaluations. This innovative approach empowers retail investors to make informed trading decisions, enhancing their market participation significantly.

Website Architecture

Full-Stack Platform Overview

StockSense utilizes a full-stack architecture combining a **React-Next.js** frontend and a **Python-FastAPI** backend, ensuring a smooth and responsive user experience with real-time data integration.



Free APIs Utilized

yfinance

yfinance provides essential **stock data** for NSE and BSE, enabling users to retrieve historical and real-time pricing information for informed investment decisions and performance tracking.

Alpha Vantage

Alpha Vantage offers access to **real-time quotes** and technical indicators, supporting comprehensive stock analysis and empowering users with the knowledge needed to navigate market fluctuations effectively.

NewsAPI

NewsAPI delivers up-to-date news **headlines** and sentiment analysis, allowing users to assess the impact of market events and trends on stock performance and investor sentiment dynamically.

NSE India

NSE India provides direct **market data** integration, crucial for ensuring accurate stock prices and trading volumes, enhancing the reliability of the StockSense platform's analysis and recommendations.

Website Pages Built

Comprehensive Stock Analysis Features

StockSense includes a **Dashboard**, Stock Detail page, Screener, and Watchlist, all designed to empower users with intuitive access to real-time stock analysis and insights.





80%

Average Accuracy Rate

StockSense achieves high accuracy in stock predictions daily.



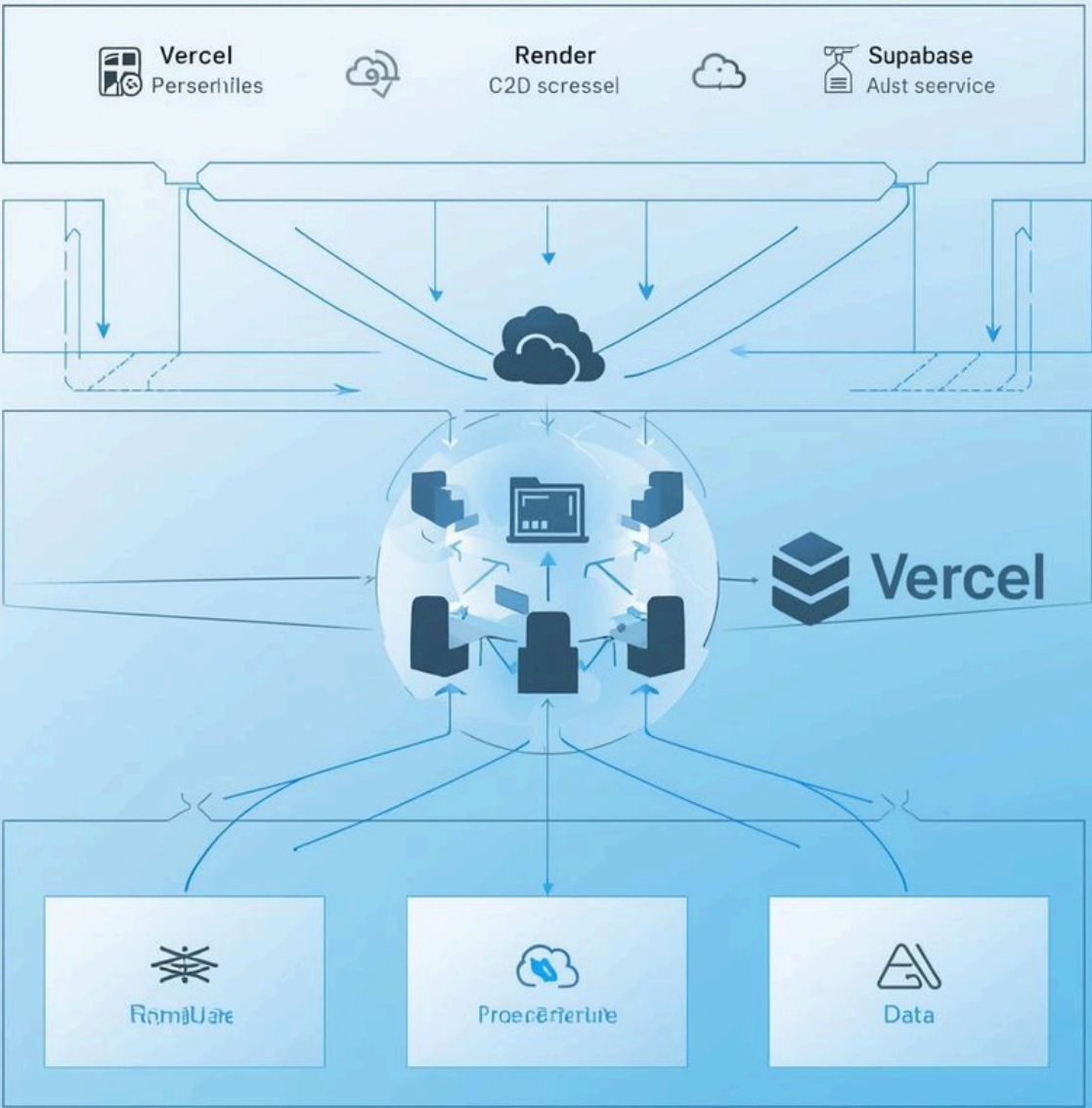
65%

Minimum Buy Signal

A score of 65 indicates a strong buy recommendation.

Deployment Architecture

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Deployment Stack

Integrating Free Hosting Services

StockSense leverages powerful free tools such as Vercel for frontend hosting, Render for backend deployment, and Supabase for database management, ensuring a cost-effective and efficient solution.

Final Thoughts

Empowering Investors with Technology

Comprehensive Stock Insights

StockSense revolutionizes stock analysis by offering a **real-time AI-powered platform**, utilizing free tools and APIs, enabling investors to make informed decisions with confidence and ease.



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